

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)
FULL SYLLABUS
26-07-2021
JEE(Main+Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCHES

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
 - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in JEE(Main) 2021

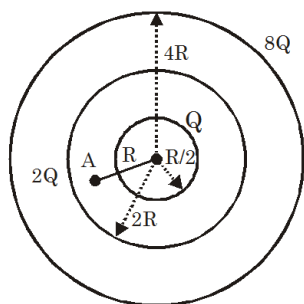
Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

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SEC-1 : (Maximum Marks: 80)

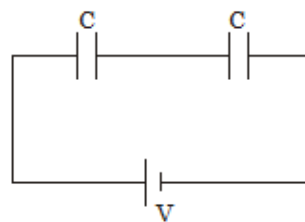
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:
Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : -1 If wrong option is selected.

- If there is an error of 1% in calculation of mass of disc & 1.5 % error in radius, then % error in moment of inertia about an axis tangent to disc is :
 (A) 2.5 % (B) 4% (C) 3.5% (D) 5%
- A particle having some charge is projected in x-y plane with a speed of 5 m/s in a region having uniform magnetic field along z-axis. Which of the following cannot be the possible value of velocity at any time ?
 (A) $3\hat{i} - 4\hat{j}$ (B) $4\hat{i} + 3\hat{j}$
 (C) $5\hat{j}$ (D) $3\hat{i} - 4\hat{k}$
- Three charged concentric nonconducting shells are given as shown in figure. Find the potential at point A :-



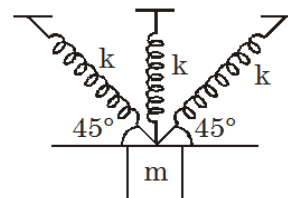
- (A) $\frac{5kQ}{R}$ (B) $\frac{4kQ}{R}$ (C) $\frac{3kQ}{R}$ (D) $\frac{2kQ}{R}$

- Initially the circuit is in steady state. Now one of the capacitor is filled with dielectric of dielectric constant 2. Find the heat loss in the circuit due to insertion of dielectric:-



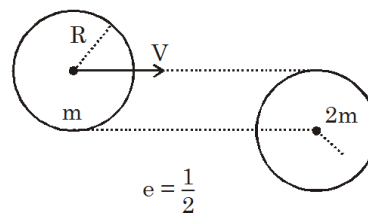
- (A) $\frac{CV^2}{12}$ (B) $\frac{CV^2}{6}$ (C) $\frac{CV^2}{3}$ (D) $\frac{2CV^2}{3}$

- Initially system is in equilibrium. Time period of SHM of block in vertical direction is :



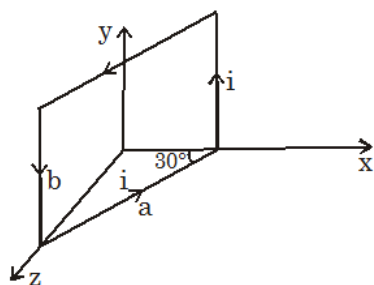
- (A) $2\pi\sqrt{\frac{m}{3k}}$ (B) $2\pi\sqrt{\frac{m}{2k}}$
 (C) $2\pi\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{2m}{k}}$

- A ball of mass m is moving with a speed V as shown in the figure. It undergoes inelastic collision with a ball of mass 2m which was initially at rest. The velocity of ball 2m after collision will be given by



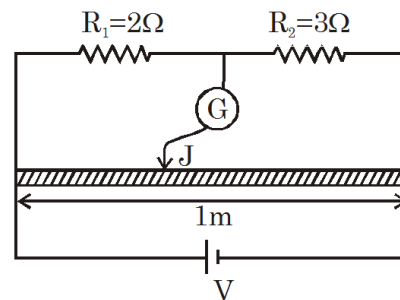
- (A) $\frac{3V}{8}$ (B) $\frac{3V}{4}$ (C) $\frac{3\sqrt{3}V}{4}$ (D) $\frac{\sqrt{3}V}{4}$

7. Current 'i' is flowing in the rectangular loop placed in the xyz plane as shown in the figure. Find the magnetic moment of the loop :-



- (A) $i(ab)\hat{k}$
 (B) $iab(\cos 30^\circ \hat{i} + \sin 30^\circ \hat{k})$
 (C) $iab(\sin 30^\circ \hat{i} + \cos 30^\circ \hat{k})$
 (D) $iab(\sin 30^\circ \hat{i} + \cos 30^\circ \hat{j})$
8. To convert galvanometer into ammeter, shunt of $0.01\ \Omega$ is used. Resistance of galvanometer coil is $50\ \Omega$ and its maximum deflection current is 20 mA. Range of ammeter is :
 (A) 0-1 A (B) 0-10 A
 (C) 0-100 A (D) 0-1000 A
9. If the maximum and minimum voltage of an AM wave are $V_{\max}$ and $V_{\min}$ then modulation index is given by
 (A) $\frac{V_{\max}}{V_{\min}}$ (B) $\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$
 (C) $\frac{V_{\max} + V_{\min}}{V_{\max} - V_{\min}}$ (D) $\frac{V_{\max}^2}{V_{\max}^2 - V_{\min}^2}$
10. Unpolarized light falls on two polarizing sheets placed one after another. What should be the angle between pass axis of the sheets so that final transmitted light have one fourth intensity of incident light?
 (A) 30° (B) 0°
 (C) 60° (D) None of these

11. In a meter bridge experiment, initially the jockey is at null point. Now resistance R_1 & R_2 is interchanged. Shift in the position of jockey is :

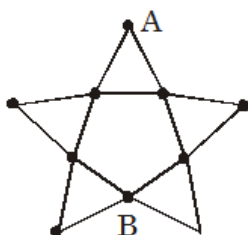


- (A) 20 cm (B) 30 cm (C) 40 cm (D) 50 cm
12. Moment of inertia of a uniform hollow hemisphere about given axis is I_A & I_B then:

 (A) $I_A > I_B$ (B) $I_A = I_B$
 (C) $I_A < I_B$ (D) Can't be predicted
13. A hydrogen atom at rest emits a photon during its transition from $n = 2$ to $n = 1$. Choose the incorrect statement.

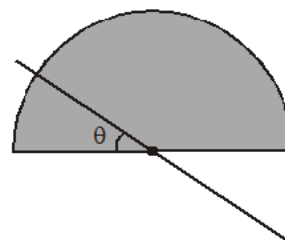
- (A) De-Broglie wavelength of hydrogen atom during recoil will be same as wavelength of emitted photon.
 (B) Energy of emitted photon will be less than 10.2 eV
 (C) Kinetic energy of recoil H-atom will be less than 10.2 eV
 (D) Kinetic energy of recoil H-atom will be equal to energy of emitted photon.

14. Resolving power of a microscope can NOT be increased by :
- (A) Using "oil immersion objective"
 (B) Decreasing the wavelength of light used
 (C) Decreasing the focal length of objective, keeping aperture same.
 (D) Decreasing the aperture, keeping focal length same.
15. If resistance between each dot is R then equivalent resistance between A & B is



- (A) $\frac{7R}{6}$ (B) $\frac{7R}{3}$
 (C) $\frac{7R}{4}$ (D) $\frac{7R}{5}$
16. If electronic charge on electron alone is doubled then as per bohr model, K.E. of an e^- revolving in the n^{th} orbit becomes
- (A) Remains same (B) Becomes 2 times
 (C) Becomes 4 times (D) Becomes 16 times
17. A monoatomic gas is taken through a process $TP^{-1/3} = \text{constant}$. If heat is given to the gas :
- (A) Its temperature remains constant
 (B) Its temperature will increase
 (C) Its temperature will decrease
 (D) Can't be predicted

18. When a load W is hung from a wire of length $2L$, it just breaks. Now this wire is completely melted and a new wire of length L is formed. If the load W is hung from this new wire :
- (A) It will definitely break
 (B) It will definitely not break
 (C) cannot be predicted
 (D) None of these
19. Two balls in free space are colliding with each other. Which of the following statement regarding linear momentum conservation of the system is true ?
- (A) is conserved
 (B) is not conserved
 (C) Will depend on type of collision (head on or oblique)
 (D) Will depend on type of collision (elastic or inelastic)
20. Moment of inertia of a semicircular disc of mass M & radius R about the shown axis is:



- (A) $\frac{MR^2}{2}$ (B) $\frac{MR^2}{4}$
 (C) $\frac{MR^2}{2} \sin^2 \theta$ (D) $\frac{MR^2}{2} \cos^2 \theta$

SEC-2 : (Maximum Marks: 40)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

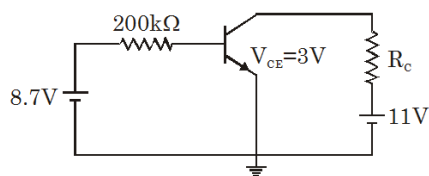
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

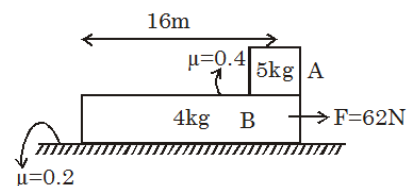
Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

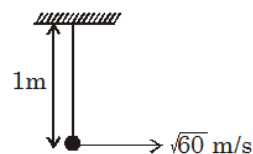
1. In a single slit diffraction pattern, if first maxima of light of wavelength λ coincides with first minima of yellow light (540 nm), then λ (in nm) is :
2. A bomb projected from ground at an angle θ ($\theta \neq 90^\circ$) explodes into two fragments of equal mass at topmost point of its trajectory. If one of the fragment returns to point of projection then ratio of De-broglie wavelength of second fragment just after explosion to bomb just before explosion is :
3. In the shown common emitter amplifier circuit, $\beta = 80$, $V_{BE} = 0.7$ volt. The value of R_C (in $k\Omega$) is :



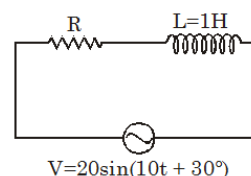
4. Initially the whole system is at rest and now a force of 62 N is applied on the block B as shown in the figure. Find the time taken (in sec) by 'A' to fall from the block 'B' :-



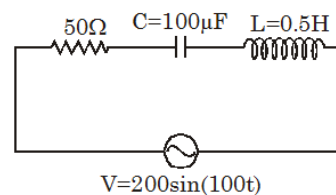
5. A stone of mass 1 kg tied on one end of the light string is whirled in vertical circle as shown in figure. Tension (in N) in the string when the string becomes horizontal is :



6. Maximum power loss (in W) across the circuit can be :



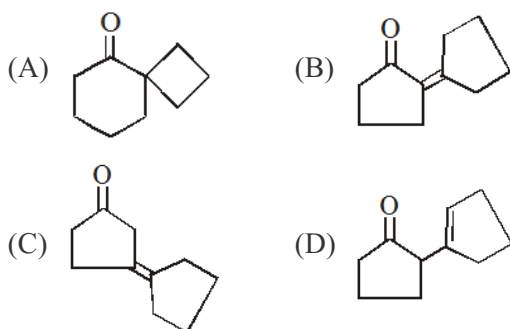
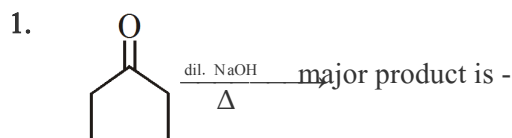
7. Brakes are being applied in a car moving with speed 20 m/s which produces a retardation of 5 m/s². The distance travelled by car (in m) till it stops is :
8. A particle is revolving in a circle of radius 2m with angular velocity $\omega = t^2 - 4t + 8$ rad/s. The time (in sec) when speed of the particle becomes 8 m/s is :
9. In the given LCR AC circuit, the effective current (in A) flowing through the circuit will be:



10. A string of length 1m fixed at both ends is vibrating in 3rd overtone. Tension in string is 200 N and linear mass density is 5 gm/m. Frequency (in Hz) of these vibrations is :

SEC-1 : (Maximum Marks: 80)

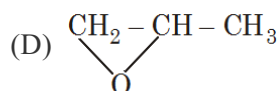
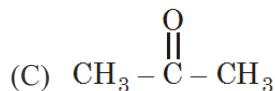
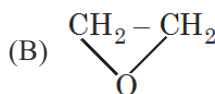
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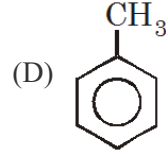
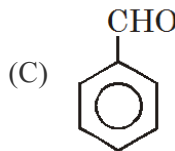
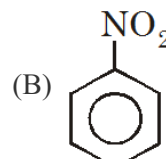
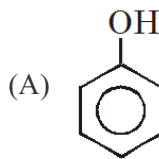
2. The correct electron affinity order is ($EA = -\Delta H_{EG}$) :-
 (A) $N^+ > O^+ > NO^+$
 (B) $O^+ > N^+ > NO^+$
 (C) $N^+ > NO^+ > O^+$
 (D) $NO^+ > N^+ > O^+$

3. For hydrogen like atoms, the average distance of electron from nucleus for the orbitals belonging to the 4th orbit, is minimum for :
 (A) 4s (B) 4p
 (C) 4d (D) 4f

4. Which of the following gives 2°-alcohol as a major product with CH_3MgBr followed by H_2O
 (A) CH_3CH_2-OH



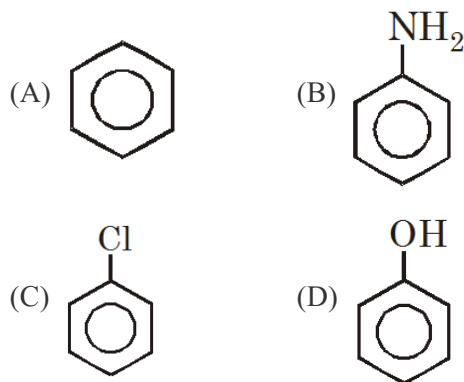
5. "On moving left to right ionisation energy increases due to increasing Z_{eff} " In which of the following pair above statement is not correct?
 (A) Na, Mg (B) Mg, Al
 (C) O, F (D) B, C
 6. The molar entropy is maximum for :
 (A) H_2O (s, 0°C, 1 atm)
 (B) H_2O (l, 0°C, 1 atm)
 (C) H_2O (g, 100°C, 1 atm)
 (D) H_2O (l, 100°C, 1 atm)
 7. Which of the following benzene ring is most reactive towards electrophile (E^+) ?



8. Van-Arkel process and Mond's process are used for refining for which of the metals respectively.

- (A) Zr and Ti (B) Ni and Zr
(C) Ti and Ni (D) Ni and Fe

9. $\text{PhN}_2\text{Cl} \xrightarrow[\Delta]{\text{H}_2\text{O}}$ product of the following reaction is?



10. $\text{CaCO}_3 \xrightarrow[\text{(Aq. suspension)}]{\text{AcOH}} \xrightarrow{\text{Na}_2\text{C}_2\text{O}_4 \text{ solution}}$

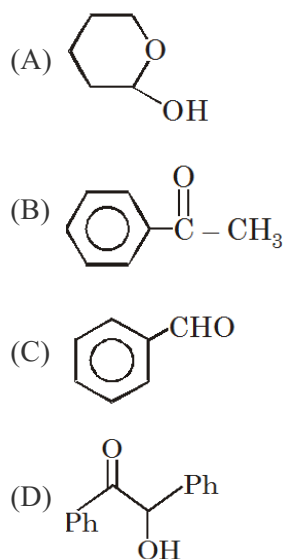
Which of the following observation is correct for the above experiment?

- (A) CO_2 is evolved
(B) Finally clear solution is obtained which is colourless
(C) Colour of initial substance and final ppt (if any) is same
(D) (A) and (C) both are correct
11. Which of the following vitamin is water soluble?
- (A) Vitamin-C
(B) Vitamin-D
(C) Vitamin-K
(D) Vitamin-E

12. Which of the following statement is incorrect?

- (A) Lu(OH)_3 is more basic compared to La(OH)_3 .
(B) Lanthanide carbide (LnC_2) produces acetylene on hydrolysis
(C) CeS is used to make Crucible
(D) Misch-metal is used for making Cigarette lighter

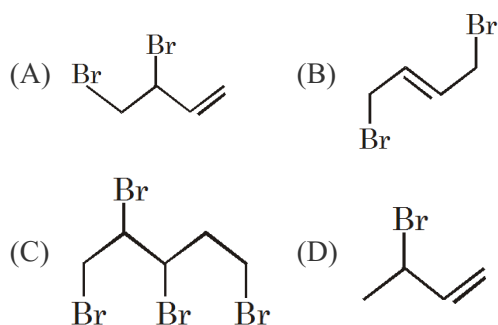
13. Which of the following compound does not give Tollen's test?



14. The only INCORRECT statement for cubic metallic crystal is -

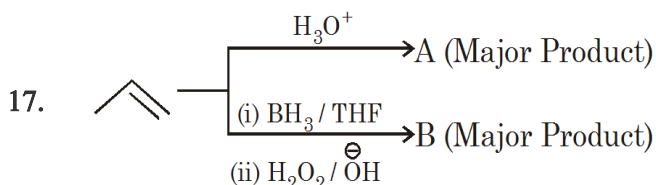
- (A) Packing efficiency (or fraction) is maximum for FCC.
(B) Co-ordination number is minimum for simple cubic (SC)
(C) For same edge-length of unit cell, the atomic radius is minimum for SC
(D) For the same atomic radius, the edge length of unit cell is minimum for SC

15.  + Br₂ $\xrightarrow{\Delta}$ (1 eq) major product is ?



16. Which of the following is electron precise type of hydride.

- (A) GeH₄ (B) CaH₂
 (C) NH₃ (D) CuH



Relation between products A and B is ?

- (A) Functional isomer
 (B) Metamers
 (C) Position isomer
 (D) Geometrical isomer

18. Which of the following compound cannot show co-ordination isomerism?

- (A) [Pt(NH₃)₄][PtCl₄]
 (B) [Co(NH₃)₄Cl₂][Cr(NH₃)₂(C₂O₄)₂]
 (C) [Pt(NH₃)₄Cl₂][Pt(SCN)₂Cl₂]
 (D) [Zn(en)₂][Zn(OH)₄]

19. Optically inactive amino acid is ?

- (A) Glycine
 (B) Alanine
 (C) Proline
 (D) Histidine

20. NaHSO₃ is heated to produce H₂O and compound 'A'. Which of the following statement is incorrect about 'A'.

- (A) A has S–O–S type of linkage
 (B) 'S' atoms have sp³ hybridisation
 (C) 'A' has S–S linkage
 (D) One 'S' has one lone pair

SEC-2 : (Maximum Marks: 40)

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The answer to each question is a Numerical Value Type questions.

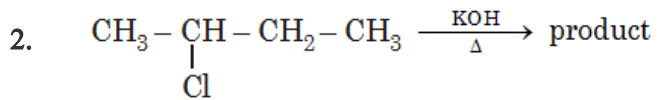
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Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 In all other cases.

1. 60 ml NaOH solution is required for complete neutralisation of 0.98 gm orthophosphoric acid. The concentration of NaOH solution is :

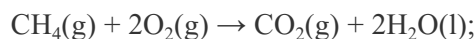


Possible number of elimination products (alkene) of the given reaction is?

3. If the number of 3c-2e bond and 2c-2e bond present in $\text{Be}_2(\text{CH}_3)_4$ are 'X' and 'Y' respectively. Give the value of (X + Y)

4. A gaseous mixture contains equal masses of dihydrogen gas, dioxygen gas and methane gas at a total pressure of 380 torr. The partial pressure of methane gas (in torr) is :

5. For the reaction :

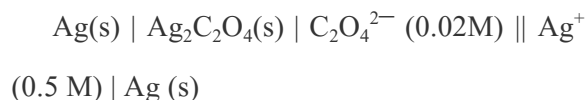


At 300 K : $\Delta_r H^\circ = -212.7 \text{ kcal/mole}$.

Find $\Delta_r U^\circ$ (kcal/mole) for this reaction at 300K.

6. For a gaseous reversible reaction, enthalpy of reaction at constant pressure is 1.8 Kcal/mol greater than that of constant volume at 300 K. The value of $\left(\frac{K_P}{K_C}\right)$ for the reaction at $\left(\frac{1}{0.00821}\right)$ K is. (Take unit of pressure & volume as 'atm' and 'litre' respectively)

7. For the Galvanic cell :



EMF of cell is 0.264 V at 25°C. If the K_{sp} of $\text{Ag}_2\text{C}_2\text{O}_4(\text{s})$ is $X \times 10^{-12}$, give X

[Given :

$$\frac{2.303 \times 8.314 \times 298}{96500} = \frac{2.303RT}{F} = 0.06, \log 2 = 0.3]$$

8. For an element the successive ionisation energy values (in eV/atom), are given below.

14.534, 29.601, 47.448, 77.472, 97.888, 552.057, 667.029

Find the number of valence shell electrons in that element.

9. The boiling point (in °C) of 0.2m- aq. urea solution is (K_b of water = $0.50 \text{ K.kg.mol}^{-1}$)

10. For a first order reaction, the rate of reaction is $2.4 \times 10^{-3} \text{ mol l}^{-1} \text{ s}^{-1}$ at 27°C. The activation energy of reaction is 24.942 kJ/mol. The rate of reaction (in $\text{mol l}^{-1} \text{ sec}^{-1}$) at 327°C is- [Take $e^5 = 150$, $e^{0.005} = 1$, $e^4 = 55$]

SEC-1 : (Maximum Marks: 80)

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Negative Marks : -1 If wrong option is selected.

- Let R_1 and R_2 be two relations on a set A , then choose incorrect statement
 - If R_1 and R_2 are transitive, then $R_1 \cap R_2$ is also transitive.
 - If R_1 and R_2 are reflexive, then $R_1 \cup R_2$ is also reflexive.
 - If R_1 and R_2 are symmetric, then $R_1 \cap R_2$ is also symmetric.
 - If R_1 and R_2 are equivalence, then $R_1 \cup R_2$ is also equivalence.
- The range of 'a' for which roots of $x^2 - 2x - a^2 + 1 = 0$ lie between the roots (exclusive) of the equation $x^2 - 2(a+1)x + a(a-1) = 0$, is
 - $\left(-\frac{1}{3}, 1\right)$
 - $\left(-\frac{1}{3}, -\frac{1}{4}\right)$
 - $\left(-\frac{1}{4}, 1\right)$
 - $\left(-\frac{1}{3}, \infty\right)$

- Let A be a square matrix such that $a_{ij} \in \{-1, 0, 1\} \forall i, j$ and it has only one non-zero entry in each row as well as in each column, then
 - A can be singular matrix
 - A must be skew symmetric
 - A must be symmetric
 - A must be orthogonal
- Consider system of equations

$$\begin{aligned} x + y - az &= 1 \\ 2x + ay + z &= 1 \\ ax + y - z &= 2 \end{aligned}$$
 - for $a \neq 1$ system has unique solution.
 - if system has no solution then 'a' must be 1.
 - for $a \in \left\{1, \frac{-1 \pm \sqrt{5}}{2}\right\}$, system has no solution.
 - for $a = \frac{-1 \pm \sqrt{5}}{2}$, system has infinite number of solutions.
- $\sum_{0 \leq i < j \leq n} i \binom{n}{j}$ is equal to
 - $n^2 2^{n-1}$
 - $(n^2 - 1) 2^{n-1}$
 - $(n-1)^2 2^n$
 - $n(n-1) 2^{n-3}$
- Let $a_1, a_2, a_3, \dots, a_{100}$ be positive real numbers and S_k be the sum of products of $a_1, a_2, \dots, a_{100}$ taken k at a time. If $S_9 S_2 \geq \lambda a_1 a_2 \dots a_{100}$, then λ is
 - $\left(\frac{100}{2}\right)^2$
 - $(9900)^2$
 - 10^6
 - none of these

7. $\sum_{r=1}^{100} \frac{\tan 2^{r-1}}{\cos 2^r}$ is equal to
 (A) $\tan 2^{99} - \tan 1$ (B) $\tan 2^{100}$
 (C) $\tan 2^{100} - \tan 1$ (D) none of these
8. If $f(x) = \begin{cases} x[x], & 0 \leq x < 2 \\ (x-1)[x], & 2 \leq x \leq 4 \end{cases}$, where $[.]$ denotes greatest integer function, then
 (A) neither $f'(1)$ exist nor $f'(2)$ exist
 (B) $f'(1)$ exists but $f'(2)$ does not exist
 (C) $f'(2)$ exists but $f'(1)$ does not exist
 (D) both $f'(1)$ as well as $f'(2)$ exist
9. The normal to the curve $2x^2 + y^2 = 12$ at the point $(2, 2)$ meets the curve again at
 (A) $\left(\frac{22}{9}, \frac{-2}{9}\right)$ (B) $\left(\frac{-22}{9}, \frac{-2}{9}\right)$
 (C) $(-2, -2)$ (D) none of these
10. If $\int e^{x^2} \left(2 - \frac{1}{x^2}\right) dx = e^{x^2} f(x) + C$ and $f\left(\frac{1}{2}\right) = 2$, then $f(1)$ is equal to (where C is an arbitrary constant)
 (A) 1 (B) -1
 (C) 2 (D) $\frac{1}{2}$
11. The eccentricity of an ellipse whose length of latus rectum is equal to distance between its foci, is
 (A) $2\sin 18^\circ$ (B) $2\cos 36^\circ$
 (C) $\sin 18^\circ$ (D) $\cos 36^\circ$
12. The general solution of differential equation $\frac{dy}{dx} + \frac{y \ln y}{x} = \frac{y(\ln y)^2}{x^2}$ is (where C is an arbitrary constant)
 (A) $\ln y = \frac{1}{2x} + Cx$
 (B) $\frac{1}{\ln y} = \frac{1}{2x} + C$
 (C) $\frac{1}{\ln y} = \frac{1}{2x} + Cx$
 (D) $\ln y = \frac{1}{x} + Cx$
13. The number of direct common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 8x - 8y + 7 = 0$, is
 (A) 0 (B) 1
 (C) 2 (D) 3
14. $P(6, 3)$ is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal at point P intersect the x -axis at $(10, 0)$, then the eccentricity of the hyperbola is -
 (A) $\sqrt{\frac{5}{3}}$ (B) $\frac{\sqrt{13}}{3}$
 (C) $\sqrt{\frac{5}{2}}$ (D) $\frac{\sqrt{13}}{2}$
15. If line $\frac{x-1}{2} = \frac{y+\alpha}{\alpha} = \frac{z-\beta}{2}$ lies in plane $2x + y + z = 5$, then $\alpha + \beta$ is
 (A) -3 (B) 4
 (C) 6 (D) -9

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a Numerical Value Type questions.

For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

16. The equation of the plane passing through the line of intersection of the planes $x + y + z = 5$ and $2x + 3y + 4z + 5 = 0$ and perpendicular to the plane $x + y + z = 5$ is

- (A) $x - z = 10$
 (B) $x - z = 20$
 (C) $x + y - 2z = 10$
 (D) $x + y - 2z = 20$

17. Three numbers are chosen at random from 1 to 15. The probability that no two numbers are consecutive, is -

- (A) $\frac{11}{32}$ (B) $\frac{44}{91}$
 (C) $\frac{33}{64}$ (D) $\frac{22}{35}$

18. A flag is standing vertically on a tower of height 1 m. On a point at a distance 2 m from the foot of the tower, the flag and the tower subtend equal angles. The height of the flag is (in meter) -

- (A) $\frac{5}{3}$ (B) $\frac{10}{3}$
 (C) $\frac{3}{5}$ (D) $\frac{3}{10}$

19. For the equation $\cos^{-1}|x| + \cos^{-1}|2x| = \pi$, the number of real solution(s) is

- (A) infinite (B) 2
 (C) 1 (D) 0

20. The negation of the compound statement $\sim p \vee (p \vee (\sim q))$ is

- (A) $(\sim p \wedge q) \wedge p$ (B) $(\sim p \wedge q) \vee p$
 (C) $(\sim p \wedge q) \vee \sim p$ (D) $(\sim p \wedge \sim q) \wedge \sim q$

1. A function f is defined on the complex number by $f(z) = (a + ib)z$, where $a, b \in \mathbb{R}^+$. This function has the property that the f -image of any point in the complex plane is equidistant from that point and origin. If $|a + bi| = 10$, then the value of b^2 is -
2. Let A and B be 2 square matrices of order 3. If $|A| = 2$ and $2B + A = A^T - B^T$ then the value of $|A + B|$ is (Where A^T represents transpose of A)
3. If $\lim_{x \rightarrow \infty} \left(\frac{a^{1/x} + b}{c} \right)^x = d$ (non zero finite), then the value of $(b + 1)\log_a d$ is -

4. Let $f(x) = \lim_{n \rightarrow \infty} \frac{(2 \sin x)^{2n}}{3^n - (2 \cos x)^{2n}}; n \in \mathbb{Z},$
 $x \neq m\pi \pm \frac{\pi}{6}; m \in \mathbb{Z}$ and $f\left(m\pi \pm \frac{\pi}{6}\right) = 0,$
 then (where $\mathbb{Z}$ denotes set of integers)

5. The value of
 $\int_0^{\frac{\pi}{2}} \sin 8x \cot x dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \ln \left(\frac{1 - \sin x}{1 + \sin x} \right) dx$ is

equal to $n\pi$, then the value of n is

6. P_1 and P_2 are two distinct and intersecting planes. Three non-collinear points lie on P_1 and another three non-collinear points lie on P_2 (none being on line of intersection of planes). Then the maximum number of tetrahedrons formed using these six points, is -

7. Area of the triangle formed by the lines $y^2 - 9xy + 18x^2 = 0$ and $y = 9$, is
8. The area bounded between the parabola $y^2 = 4x$ and the line $2x + y - 4 = 0$ is
9. Let $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} + 2|\vec{c}| = 0$ and $|\vec{c} - \vec{a}| = \sqrt{14}$ and angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ is 30° , then $|(\vec{a} \times \vec{b}) \times \vec{c}|$ is
10. The mean of 10 terms is 3. If the first term is increased by 1, second by 2 and so on, then the new mean is -

Space for Rough Work